

**TECHNOTHLON 2020
HAUTS
THEME - THE LION KING**

SECTION A

It's a beautiful morning in the Pride Lands. Zazu the hornbill and Simba are playing a game of marbles.

QUESTION 1

Nine marbles are kept on a rock and numbered 1 through 9. On each turn, a player takes one of the marbles off of the rock and keeps it. The first one who can make exactly three of their marbles add up to 15 is the winner. If Simba goes first, and both play optimally, who will win?

a) Simba

b) Zazu

**c) Always end
in a draw**

d) Can't Predict

The King, Mufasa takes Simba with him on a walk to patrol the Pride Lands. He shows him the whole kingdom and tells Simba that someday, he will be the one ruling it and protecting the animals. While talking to Simba about the Circle of Life, they come across a field.

QUESTION 2

The field is divided into 25 squares, with numbers written on each square as shown in the figure. Mufasa thinks that it is a good puzzle for Simba to put them all into regular order so that the first line reads 1 2 3 4 5, and the second line reads 6 7 8 9 10, and so on to the end, by taking up one counter in each hand and making them change places. Thus, Simba might take up 7 and 1 and replace them as 1 and 7. Then take up 24 and 2 and make them also change places, then Simba will have the first two counters properly placed. The puzzle is for Simba to determine the fewest possible exchanges in which this can be done.

5	3	17	18	12
9	13	1	25	23
6	19	20	2	21
16	14	15	8	7
4	24	10	22	11

A. 17

B. 22

C. 20

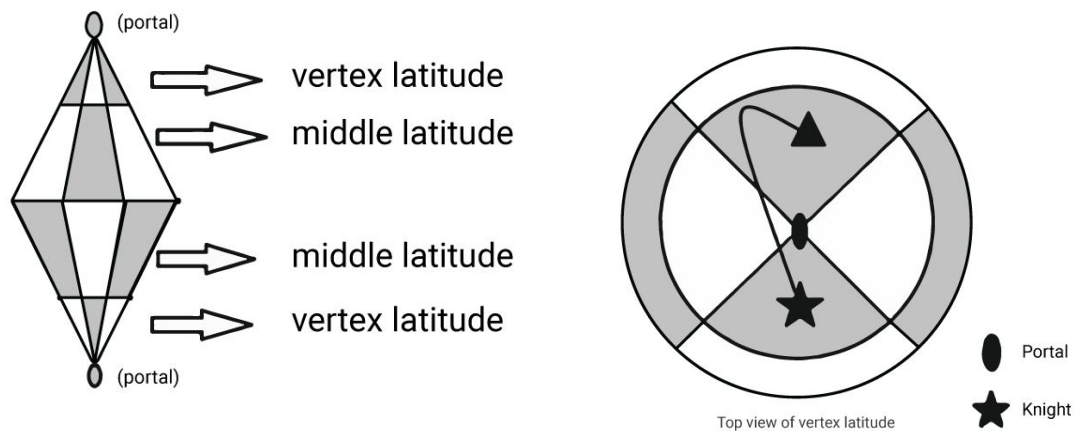
D. 18

Hearing some commotion near the fields, Mufasa sends Simba back to the cave with Zazu. On their way, they meet the other lion cubs and walk by Scar's cave, who is Mufasa's younger brother and covets the throne and is plotting to eliminate Mufasa and Simba to become King.

For questions 3 and 4

To test Simba and gauge his intelligence, Scar asks Simba a riddle. Scar says "Imagine my cave as two mountains (cones) joined together, such that a 4*4 chessboard is formed on it, as shown in the figure. Simba and his friends behave as

knights of chess. Let x be the minimum number of knights required to cover the whole board and y be the number of unique patterns using exactly x knights with at least one knight in either of the vertex latitudes and at least one knight in either of the middle latitudes. (Remember that each pattern should be different even if the cone is rotated along a vertical axis.)



RULES FOR MOVEMENT

- 1) The knights move in the same fashion as in the normal chessboard. (Only $2x$, $1y$ & $2y$, $1x$ movement is replaced with 2θ , $1r$ & $2r$, 1θ movements respectively.)
- 2) Using the upper/lower vertex as a portal, one can bring the knight to the other extreme latitude (vertex latitude) of the same longitudinal file.
- 3) The knights can also jump over the portal like in fig. 2 and the jump would be considered as a radial move only. (radial means movement along the longitudinal file)

RULES FOR COVERING THE BOARD

- 1) To capture a vertex square, it should either be attacked by at least one vertex knight or two middle knights, or by both one vertex knight and two middle knights.
- 2) To capture a middle square, it should either be attacked by at least one middle knight or two vertex knights, or by both one middle knight and two vertex knights.
- 3) If a knight is on a certain square then, that square is considered to be completely attacked.

QUESTION 3

What is the value of x ?

a) 3

b) 4

c) 5

d) 6

QUESTION 4

What is the value of y ?

a) 10

b) 8

c) 12

d) 16

QUESTION 5

Mufasa takes Simba to visit a nearby village in the evening, where a festival is being organized. There are many lights and fireworks. They see two villagers operating the laser-like lights. The villagers leave, and Simba and Mufasa decide to have some fun. Suppose they have an unlimited number of laser nodes that can emit a green or blue light between any two nodes. They can't control which color any pair of nodes will

produce. In order to make an efficient light source, they need to have at least one polygon of rays of the same color (e.g. an all green square or an all-blue triangle).

What is the smallest number of nodes among the following needed to ensure that a light source is guaranteed to be formed? (note: the nodes are always arranged as vertices of a regular polygon.)

a) 5

b) 6

c) 7

d) 8

SECTION B

QUESTION 6

It's dinner time, and Mufasa has hunted a deer for the tribe. Simba and his friend Nala divide a triangular piece between themselves. Simba chooses any point in the piece and Nala makes a straight cut through this point and chooses the part. Find the size of the part that each of them can guarantee for himself/herself for Simba and Nala respectively (both of them want to get as much as possible).

a) $5/9, 4/9$

b) $4/9, 5/9$

c) $1/2, 1/2$

d) $1/3, 2/3$

Scar tricks innocent Simba and Nala into exploring the forbidden elephants' graveyard, which is the home to numerous hyenas, thinking that the hyenas will kill Simba and Scar will be one step closer to the throne.

QUESTION 7

While heading towards the elephants' graveyard, Simba and Nala come across a cave opening in the shape of a square wall of sides 1 km, with 4 holes. Nala and Simba need to cut a square in order to get through the wall. Which of the following is the largest length (in m), such that they can always cut a square of that length's sides without having a hole in its area? (Assume the holes to be of point size and they can be anywhere on the wall)

a) 589

b) 486

c) 369

d) 360

For question 8 and 9

Nala and Simba enter the graveyards and are attacked by the dangerous hyenas. The hyena queen thinks of a clever way of trapping them. She traps them in a dark cave with n number of magical wells. The water in $n - k$ number of wells is poisoned and the rest k are poisonless. Simba knows which wells are poisoned and which are poisonless. Since Simba is the son of King Mufasa, he can pick a well, and make a wish to reverse the configuration of all the wells(i.e. poisoned to poisonless and vice versa) except for the one he chose.

For a particular value of n , if suppose x is the minimum number of times he needs to make a wish so that in the end the maximum number of poisonless wells are present for a particular value of k . And let y is equal to the sum of all x as k varies from 0 to n ($0 \leq k \leq n$) for a particular value of n .

QUESTION 8

Find the sum of all y for all the even values of n from 0 to 20? (both inclusive)

a) 825

b) 775

c) 245

d) 325

QUESTION 9

Then find the sum of all y for all the odd values of n from 0 to 20? (both inclusive)

a) 775

b) 294

c) 245

d) 754

Mufasa reaches the graveyard in time and is able to save Nala and Simba. After the failed attempt of the hyenas to kill Simba, Scar pays them a visit at night and tells them his plan to kill Mufasa and Simba, telling the hyenas how the kingdom will be his and they will be able to do as they please after they both are gone. The hyenas don't trust Scar and ask him a riddle to test his wits.

QUESTION 10

On a big rock, Shenzi writes the following expression.

$$10 \div 9 \div 8 \div 7 \div 6 \div 5 \div 4 \div 3 \div 2$$

Shenzi tells Scar that he must insert brackets in the written expression in such a way that the resulting number is an integer. The riddle does not end here. She asks him to determine the maximal integer value as well as the minimal integer value of this expression, say A and B. Find $A - B$ and give the last digit of the sum of digits of that number.

a) 5

b) 9

c) 7

d) 3

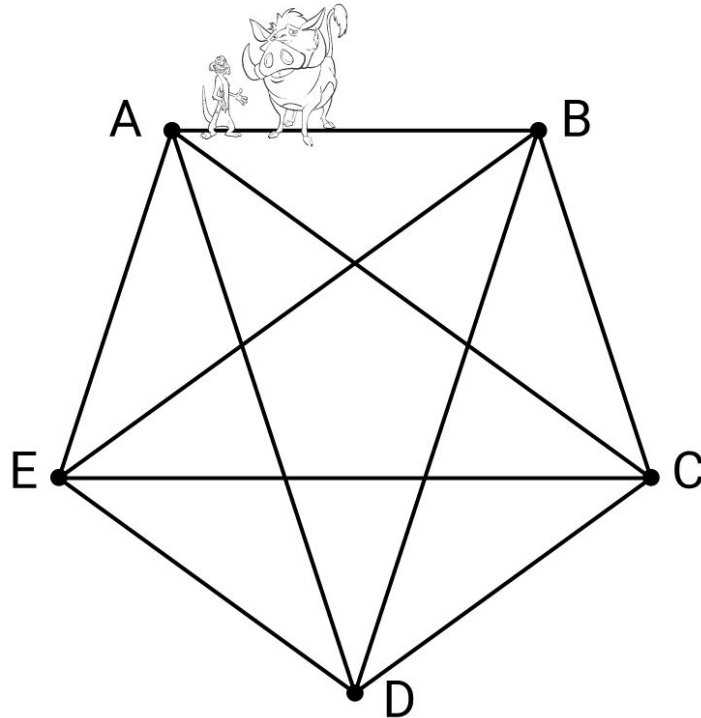
Scar is able to succeed in his plan and kills Mufasa by pushing him over a cliff. Simba, on the other hand, also falls but lands on a tree and is presumed dead by the hyenas. He leaves the kingdom thinking Mufasa's death was his fault. Scar becomes the King and allows Shenzi's clan to live in the Pride Lands.

SECTION C

Tired of walking across the desert for days with no food and water, Simba collapses on the sand and is found by Timon and Pumbaa who take him to their home, The Oasis.

QUESTION 11

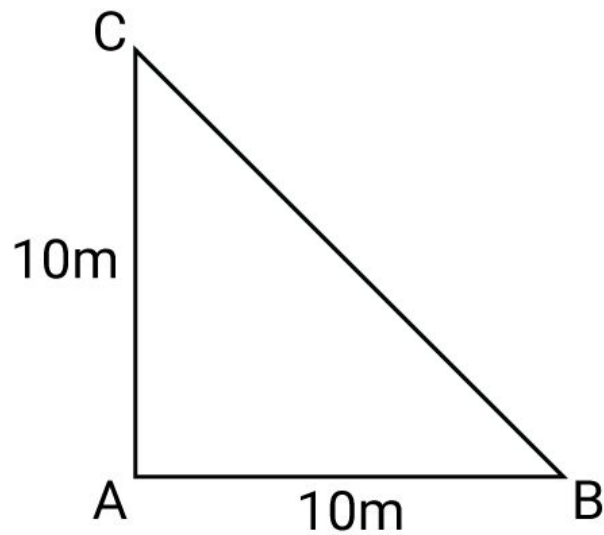
Timon and Pumbaa become Simba's friends. Simba loves how beautiful The Oasis is, and decides he wants to roam around the whole forest. Let A, B, C, D, and E be the key places they visit in their tour, which starts from point A. They wish to make a complete tour of these roads, going along every one of them exactly once and reaching block A in the end. How many different routes are there from which they can select?



It has been a few months since Simba left the Pride Lands, and now a young adult he lives his life with his new family in the Oasis.

QUESTION 12

The Oasis animals decide to get some exercise and end up playing a special race. The arena is in the shape of an isosceles right-angled triangle ABC (hypotenuse BC) of base sides of length 10 meters. Timon starts at the vertex A. He has to touch the side BC then side AC then side AB then side BC again then side AB again and finally reach the vertex C (see figure). What is the shortest distance that he has to travel? Give an answer rounded off to the closest integer.



QUESTION 13

Nala wants to escape the Pride Lands which is now ruled by Scar and the hyenas. The only problem is that there is only one way out of the Pride Lands, which is guarded by a hyena and requires a special number. Zazu overhears the code and decides to tell Nala, but is captured by the hyenas. The hyenas test Zazu to see if he really knows the special number, and ask him some questions. Little do they know that Nala is hiding in the bushes hearing the conversation. Nala can only tell whether Zazu's saying the truth or lying (she has this particular deductive skill). The hyenas are trying to get out the code which lies between 11 and 1111 (both inclusive). They ask the following set of questions.

1. Is the number less than 500?

Zazu lies

2. Is the number a perfect square?

Zazu lies

3. Is the number a perfect cube?

Zazu says the truth

4. Is the second digit 1

Nala couldn't make out whether he lies or not.

The hyenas are convinced that Zazu knows the wrong code and let him leave. However, Nala successfully deciphered the code. What is it?

QUESTION 14

Feeling bored in the afternoon, the Oasis animals decide to play a game that Pumbaa invented. In this game ' n ' animals stand around in a circle, and Pumbaa tells them to write a value, of either $+1$ or -1 . He then performs the following operation as many times as he wants: between all the consecutive numbers on the circle, he writes their product, and the initial n numbers are deleted. Suppose that for some n for any initial arrangement of $+1$ and -1 on the circle, after finitely many operations all the numbers on will be equal to $+1$. What is the sum of all such n lying between 1 and 345 (included)?

QUESTION 15

The Oasis animals get to know about the coronavirus pandemic from a nearby village. They visit the village to check the situation. They find that the fight against the deadly coronavirus has come to a final showdown. All the virus in the world has been swept away, except one. However, the bad news is that this final virus has mutated to the level where one can never kill it, but only trap it forever. The villagers have successfully trapped the virus in a 5x5 maze (each block =1 cmx1cm) from which they cannot afford to let it escape. The virus resides at the centre of the block and can only move from the centre of one block to another block. The virus has developed a function according to which you need to introduce it to n chemicals ... which will denote the distance(in cm) it can move in a turn. The numbering of each chemical denotes the distance the virus must travel in a particular minute and the number on each chemical must be **distinct**.

The chemical to be used is decided by the virus and the direction of movement is decided by us (not necessarily horizontal or vertical). We need to create a set of numbers with max possible no. Of elements, S , such that no matter which no. The virus chooses from that set, we can make sure it doesn't escape the maze ever.

Each minute the virus will be positive for a particular chemical (this is random, the villagers have got no control over it), and they must move it to another block, the distance corresponding to the chemical. What is the max value of n and what are the n chemicals i.e. the distances induced by these chemicals(in cm) that one must choose, given that the virus begins at the top left room? State your answer as the sum of the chemical numbers of set S . If it's impossible to find such n chemicals, state the answer as 0.

Note(all distances moved must always be integral)

SECTION D

QUESTION 16

Simba and Nala head back to the Pride Lands to dethrone Scar, and on their way, they come across a barren field of dimensions $m \times (m \times k)$, where m is the number of rows and $(m \times k)$ is the number of columns. Suppose Simba and Nala use some tiles covered with seeds to cover the barren land. Simba uses tiles of the order $1 \times m$ & Nala uses tiles of the order $m \times 1$. For each tile they place on the field, a plant grows on the tile. Both players place their tiles on the field one by one, if at some point a player is not able to place his tile on the field, then he loses and the opponent gets ten more plants to grow. And after one player loses, as the winning opponent has some extra places to place his tile then he will continue to place the tiles until there are no more places left. Nala starts by default. Given that $2 \leq m \leq 10$ and $10 < k \leq 100$.

After they play all the games possible with different values of m and k under the given range, find the difference in the number of plants they both grew. (Assume that both the players play optimally)

a) 2025

b) 1000

c) 8100

d) 1025

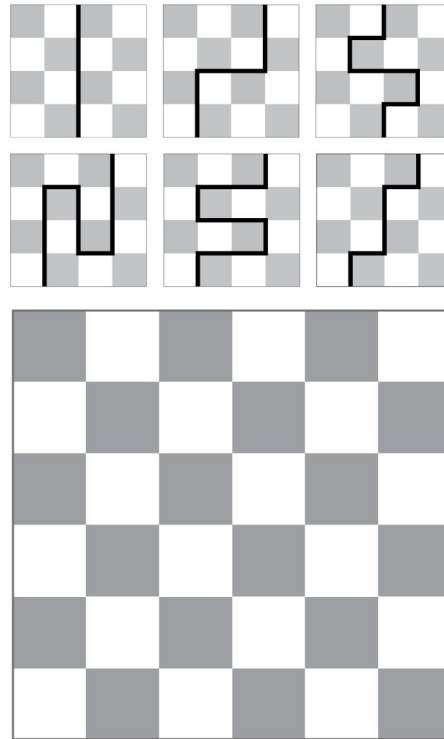
QUESTION 17

Timon and Pumbaa also decide to go help their friend Simba and figure out a shortcut to reach the Pride Lands quickly. They pass through a very dense forest, which is in the shape of a chessboard. Their journey through the forest is a long one, and both of them try to figure out how many different ways may a chessboard be divided into two parts of the same size and shape by cuts along the lines dividing the squares?

Note - Two shapes are considered the same if one can be transformed into another by some number of 90-degree rotation or flipping.

Example - We take a board of smaller dimensions. It is obvious that a board of four squares can only be so divided in one way—by a straight cut down the centre—because we shall not count reversals and reflections as different. In the case of a board of sixteen squares—four by four—there are just six different ways. All the

cases are given in the diagram, and the reader will not find any others. Suppose the forest is a larger board of thirty-six squares. So, in how many ways may it be cut into two parts of the same size and shape?



a) 216

b) 280

c) 170

d) 255

QUESTION 18

Simba, Nala, and Timon and Pumbaa meet at the Pride Lands and see the hyenas guarding the entrance. The hyenas, Banzai and Ed are playing a game- There are two boxes A and B with m balls in one box and n balls in another. A player out any number of balls from any one of the boxes or you can take out an equal number of balls from both the boxes. For example, the player can take out 4 balls from A or 7 balls from B or 3 balls from both A and B. In the end, both A and B will be empty. The last hyena to make the move wins. Banzai always insists on going first in the game. If n varies from 2 to 20 and m varies from 1 to $n - 1$ and both players play optimally then in how many games does Banzai have the winning strategy?

a) 182

b) 171

c) 95

d) 86

The hyenas spot Simba and his friends, and they all run and split up. While Simba and Nala fight off some hyenas, Timon and Pumbaa run into a cave and block the entrance with a boulder.

QUESTION 19

The cave has 16 blocks with 5 passages/corridors each in north-south and east-west directions. When they (Timon and Pumbaa) cross one block in the east direction the hyena count increases by 2 and doubles for every block passed going south. The hyena count in the cave begins at zero at the top left corner(entrance) and Timon and Pumbaa reach to the topmost row's middle point of the cave-The Main room(This is where they get to know the rules regarding the hyenas). This means the current hyena count in the cave is 4. But all hope is not lost. For every block they move north, the hyena count halves, and west - decreases by 2. Also, they cannot pass the same corridor in any direction twice. However, they can pass an intersection twice. They must get to the exit without the hyenas, which is at the bottom right corner of the lab. (Note: a negative count is possible: you can think of it as Timon and Pumbaa being able to fight off the hyenas.)

If every block has dimensions 1m X 1m then what distance will Timon and Pumbaa travel to reach the exit from the entrance, including reaching the main room from the entrance?

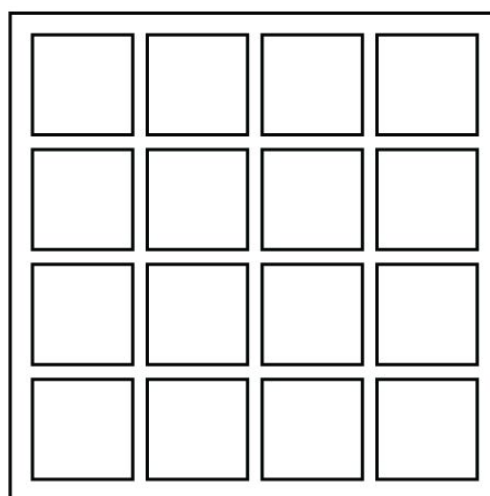
How many turns will they make? Give the sum of both.

a) 30

b) 28

c) 26

d) 24



The other lions come to Nala and Simba's help. It's the lion herd vs Scar and the hyenas.

QUESTION 20:

The final battle occurs in an interesting manner. Scar knows he can't defeat Simba, so he runs and hides in a cave. The cave was built by Simba's ancestors, and Simba must solve a final riddle to open the cave, which is as follows.

Let there be a new chess piece known as the king rook. This piece moves like a rook, but like the King, it could move only one square in the horizontal and vertical directions. There is one more constraint, that suppose in a move if the king rook moved to the left or right square then whenever it is moved again it has to move to the up or down square, and vice-versa.

Suppose there is a chessboard of the order $101 \times m$. Simba chooses an integer k from the set $\{1, 2, 3, 4, \dots, 101 \times m\}$ blindfolded. Simba places these k king rooks on the board (optimally). In every turn, Simba has to move every king rook to one square (explanation: moving each and every king rook exactly once to a valid empty square one by one will be considered as a turn. Moving one king rook is not considered as a turn, moving every king rook is considered as a turn). If after a certain number (possibly zero) of turns Simba is not able to make a complete turn then he is unsuccessful. If the turns continue forever then Simba is said to be successful. Suppose x is the value of the maximum possible k from the set so that Simba is always successful. And y is the number of squares left after placing the x king rooks (i.e. $y = 101 \times m - x$). Now, Simba must play the puzzle for each m from the following set $\{101, 103, 105, \dots, 117, 119\}$. Find the sum of all y for all the different values of m ? (In every turn Simba moves the king rooks in an optimal way).

Simba is able to crack the riddle and banishes Scar from the kingdom. The other lions are able to fight off the hyenas, letting them live with the condition that the hyenas must never come out of the graveyards. Simba becomes the king, and peace is restored in the Pride Lands.

a) 1090**b) 2090****c) 1080****d) 2100**